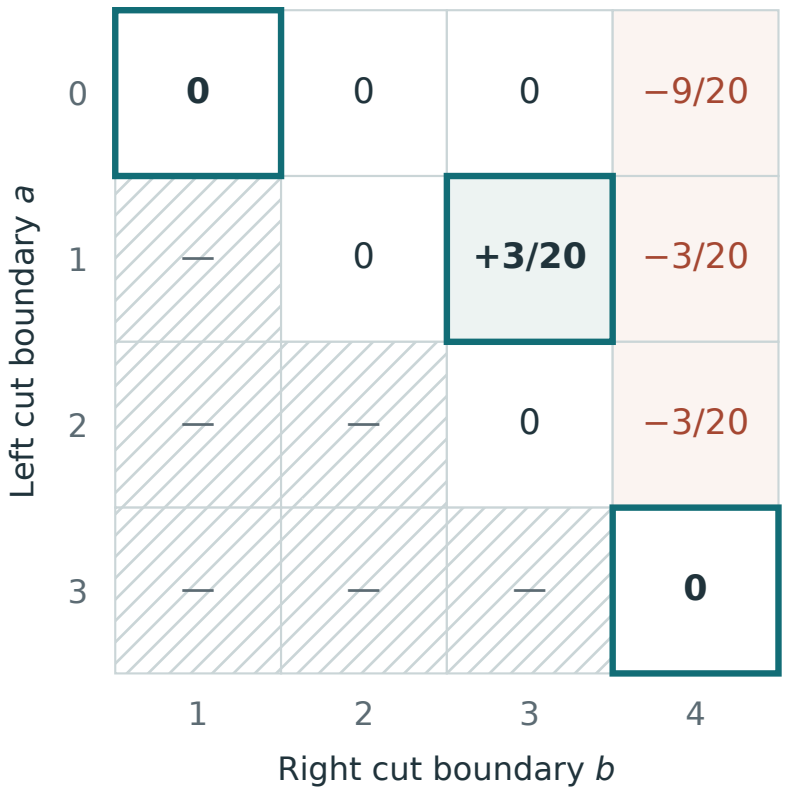
A dynamic program chooses grade boundaries

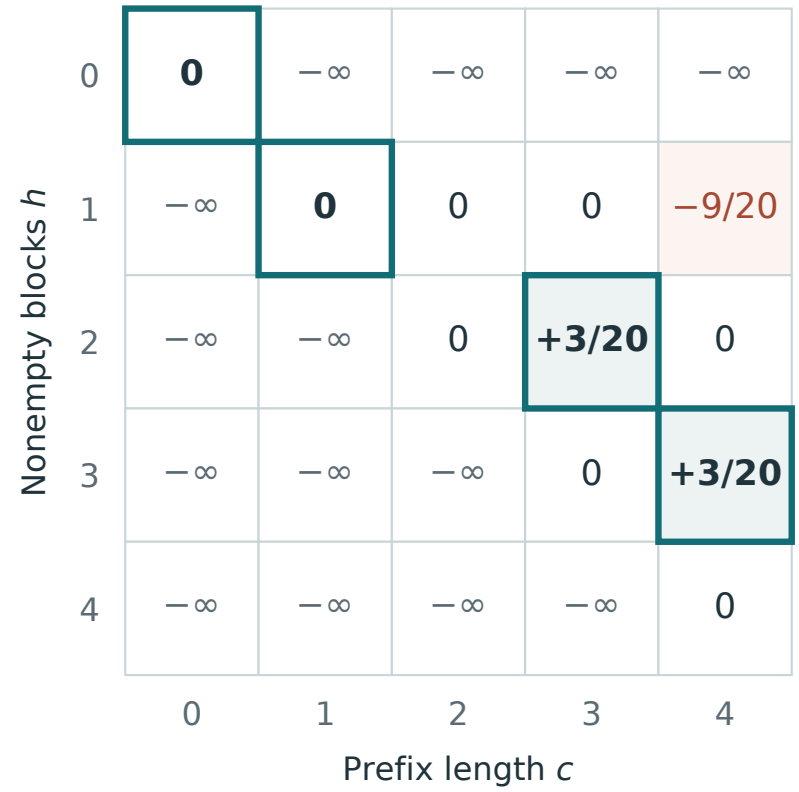
Fixed order A, B, C, D; w = (3/5, 2/5), τ = 1/4; M = 6.

(a) Signed interval savings $W(a, b)$



W(a,b) covers positions a+1 through b.
Signed savings may be negative.

(b) Prefix maximum savings $V_h(c)$

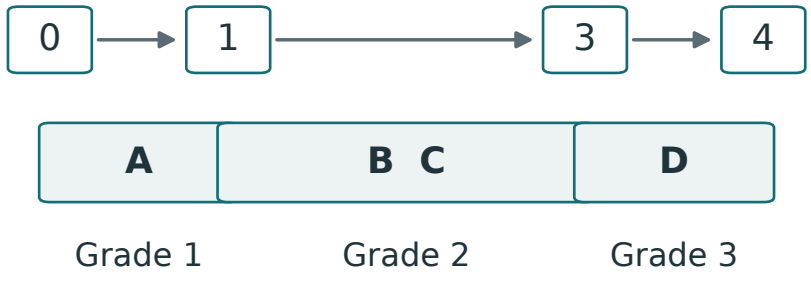


−∞ is infeasible; V₀(0)=0 is auxiliary.
Outlines select the three-block solution.

$$V_h(c) = \max_{h-1 \le b < c} \{V_{h-1}(b) + W(b, c)\}$$

The state value is an unnormalized maximum saving.

(c) Recover the three-block report



$g = (1, 2, 2, 3), \quad N = 5$

Saving 3/20; risk reduction 1/40.

$R = 7/40 - (3/20)/6 = 3/20$

Exactly 4 grades	At most 4 grades
Risk 7/40	Risk 3/20

(d) Guarded neutral extension

Separate symbolic qualification

A full active-profile equality class H must stay within one grade.

No cut may split H between its first and last positions.

Use the neutral loss and its baseline.
Keep original units and pair multiplicities.

Separate from the tie-free example;
no package DP implementation is asserted.